

Chapter 14: AI for RNA Therapeutics & Nucleic Acid Design

Chapter 13 closed by handing off to a different real molecular modality: RNA. Every chapter before this one treated nucleic acids either as a genome-scale data source (Chapter 4’s ChEMBL/PubChem/UniProt pipelines) or as the thing a small molecule or a protein binds to; none of them treated RNA itself as the therapeutic modality — the molecule that is designed, optimized, and delivered as a drug. That gap matters now for a concrete, real reason: RNA therapeutics stopped being a niche in the last decade. Small interfering RNA (siRNA) drugs (patisiran, givosiran, lumasiran, inclisiran) and mRNA vaccines both reached real regulatory approval and, in the case of the mRNA COVID-19 vaccines, real global deployment at a scale no earlier nucleic-acid therapeutic approached — built on real, specific molecular design choices this chapter covers directly. RNA is also a biophysically distinct design problem from the small molecules and proteins of Chapters 1-13: it is simultaneously the therapeutic molecule *and*, to a much greater extent than a folded protein, an object whose function is legible directly from its own sequence through base-pairing thermodynamics — a property this chapter’s own tooling (Section 14.1) and hands-on project (Section 14.3) both exploit directly, computationally, and cheaply, without the GPU-scale structure-prediction models Part III required for proteins.

This chapter covers real RNA secondary- and tertiary-structure prediction and what each is actually good for (Section 14.1); the real, established design principles behind optimizing an mRNA or an siRNA for stability, translation, and low immunogenicity (Section 14.2); and a real, complete hands-on project that predicts siRNA knockdown efficacy directly from real, published RNAi silencing data and real, self-computed RNA thermodynamics (Section 14.3).

14.1 RNA Structure Prediction

14.1.1 A different structure-prediction problem than Part III’s

Part III’s central finding — most explicitly in Chapter 9 — was that protein tertiary structure is extraordinarily hard to predict from sequence alone by first-principles physics, and that it took evolutionary information (multiple sequence alignments) and, later, enormous learned models (AlphaFold2/3, ESMFold) to close that gap. RNA presents almost the opposite profile. **RNA secondary structure** — the set of Watson-Crick ($A = U$, $G \equiv C$) and wobble ($G \cdot U$) base pairs a single RNA strand forms with itself — is, to first approximation, a direct, computable consequence of the sequence’s own nearest-neighbor base-pairing thermodynamics, solvable by exact dynamic programming in polynomial time, no learned model or evolutionary information required. This is not a minor convenience: for many of RNA’s real biological and therapeutic roles — a small interfering RNA’s guide-strand loading, a microRNA’s seed-pairing target recognition, a riboswitch’s ligand-sensing conformational switch, an mRNA vaccine’s 5’ UTR translation efficiency — the 2D secondary structure alone already captures most of what matters functionally, which is exactly why Section 14.3’s hands-on project builds an entire real predictive pipeline on secondary-structure and duplex thermodynamics without ever needing a 3D coordinate. **RNA tertiary (3D) structure** prediction, by contrast, remains a substantially harder and less mature problem than protein 3D structure prediction is today (Section 14.1.4) — the inverse of Part III’s own difficulty ordering.

14.1.2 The nearest-neighbor thermodynamic model and the Zuker MFE algorithm

A secondary structure s for a sequence of length n is a set of base pairs (i, j) , $i < j$, with no nucleotide participating in more than one pair and no two pairs “crossing” ($i < k < j < l$ for two pairs (i, j) and (k, l) — the no-pseudoknot restriction Section 14.1.3 returns to). The **nearest-neighbor thermodynamic model** (Turner & Mathews, 2010) assigns structure s a total free energy as an additive sum over its local structural elements — stacked base pairs, hairpin loops, bulges, internal loops, and multi-branch loops — each parameterized by a real, experimentally measured free-energy increment (from UV-melting experiments on thousands of short synthetic duplexes, curated into the Nearest Neighbor Database, NNDB). A stacked pair of adjacent Watson-Crick base pairs, for instance, contributes a real, sequence-context-specific ΔG around -1 to -3 kcal/mol; an unpaired hairpin loop contributes a real, loop-length-dependent *penalty* (entropic cost of constraining a loop). No physics-based first-principles calculation is involved — the model is entirely built from these curated, real, measured increments, exactly the same “trust real, published experimental measurements over a from-scratch physical simulation” strategy Chapter 6’s bond-length priors and Chapter 12’s classical force fields both already relied on.

Finding the secondary structure that minimizes this total free energy — the **minimum free energy (MFE) structure** — is the real dynamic program Zuker and Stiegler (1981) introduced (the algorithm ViennaRNA’s `RNA.fold()` still implements, in a considerably more complete and optimized form, four decades later). Two coupled recursions suffice for the pseudoknot-free case: $W(j)$, the MFE over the prefix $1\dots j$ allowing j to be unpaired or paired to any earlier position, and $V(i, j)$, the MFE over the substructure enclosed by pair (i, j) , *given that* (i, j) pairs:

$$W(j) = \min \left(W(j-1), \min_{1 \leq i < j} \{ W(i-1) + V(i, j) \} \right)$$

$$V(i, j) = \min \left(eH(i, j), \min_{i < i' < j' < j} \{ eL(i, j, i', j') + V(i', j') \}, \min_{\text{multiloop split}} \{ \dots \} \right)$$

where $eH(i, j)$ is the real, measured hairpin-loop energy for closing pair (i, j) , and $eL(i, j, i', j')$ is the real, measured stacking or internal/bulge-loop energy between an outer pair (i, j) and an inner pair (i', j') (a simple stack when $i' = i + 1, j' = j - 1$; a bulge or internal loop otherwise). Filling both matrices takes $O(n^2)$ space and $O(n^3)$ time (the internal-loop term’s own double sum, in its unoptimized form) — fast enough to fold thousands of siRNA-length sequences in well under a second each, which is precisely why Section 14.3’s project can afford to fold every one of 2,361 real sequences individually rather than relying on any pre-computed structural feature.

14.1.3 The McCaskill partition function and what it buys beyond the single MFE structure

The MFE structure is a single point estimate — the *one* lowest-energy fold — but a real RNA molecule in solution exists as a thermal ensemble over many accessible conformations, particularly in loosely structured or marginal-energy regions. McCaskill’s (1990) partition-function algorithm replaces Zuker’s min recursions with the analogous log-sum-exp recursions over the same pseudoknot-free structure space Ω , computing the real Boltzmann-weighted partition function

$$Z = \sum_{s \in \Omega} e^{-E(s)/RT}$$

and, from it, the real **base-pair probability** p_{ij} — the fraction of the thermal ensemble in which positions i and j are actually paired:

$$p_{ij} = \frac{1}{Z} \sum_{s: (i,j) \in s} e^{-E(s)/RT}$$

This is directly useful, not merely a theoretical refinement. This chapter’s own hands-on project (Section 14.3) uses it twice: the **mean base-pair distance** to the ensemble (`RNA.fold_compound.mean_bp_distance`, ViennaRNA’s real ensemble-diversity statistic) as a confidence-like feature — a guide strand whose MFE structure is close to the entire thermal ensemble is a confident, well-determined fold, while a large mean distance flags a genuinely undetermined structure, exactly the role pLDDT’s *confidence*, not *accuracy*, framing played for protein structures in Chapter 9 — and, more directly, the real target **accessibility** calculation the next paragraph describes, the single most important RNA-thermodynamic feature this chapter’s own model relies on.

Target-site accessibility. A target mRNA site that is already single-stranded in its own local secondary structure is easy for RISC (the RNA-induced silencing complex) to engage; a site buried inside a stable stem-loop is not, regardless of how good the siRNA-target duplex would be in isolation — an established, real determinant of siRNA potency RNAup (Mückstein et al., 2006) was built specifically to quantify. This chapter’s own `target_accessibility_features()` (`siRNA_efficacy.py`) computes the same real quantity RNAup does, by the most direct real route ViennaRNA’s Python bindings expose: fold the 57-nt real local genomic context once unconstrained (real MFE E_{free}), fold it again with a real hard constraint (`fold_compound.hc_add_up`) forcing every position in the central 19-nt target window to be unpaired (real MFE E_{open}), and report the real, non-negative energetic cost of opening that window,

$$\Delta G_{\text{open}} = E_{\text{open}} - E_{\text{free}} \geq 0.$$

```
def target_accessibility_features(extended_target: str) -> dict:
    fc_free = RNA.fold_compound(extended_target)
    _s_free, mfe_free = fc_free.mfe()

    fc_open = RNA.fold_compound(extended_target)
    for i in range(TARGET_WINDOW.start + 1, TARGET_WINDOW.stop + 1): # 1-indexed
        fc_open.hc_add_up(i)
    _s_open, mfe_open = fc_open.mfe()

    return {
        "target_unconstrained_mfe": round(mfe_free, 3),
        "target_opening_energy": round(mfe_open - mfe_free, 3),
    }
```

A concrete, real example from this chapter’s own cached data (Section 14.3): the real Huesken siRNA GGAAGGUGAUGCUUAUUAUU, targeting its real 57-nt local context, has an unconstrained MFE of -7.8 kcal/mol and a constrained (target-window-forced-open) MFE of -1.5 kcal/mol — an opening energy of 6.3 kcal/mol, a real, substantial accessibility penalty (its measured knockdown efficiency, on this dataset’s real $[0, 1]$ scale, is a modest 0.39). By contrast, UUAGUGAGAAUUCUGCAAC, a real siRNA measured at 0.91 efficiency, has an opening energy of only 4.9 kcal/mol — a real, less-structured, more accessible target site, consistent with, though not proof of, the accessibility hypothesis; Section 14.3’s own feature-importance results (not this one illustrative pair) are the real, quantitative test of that hypothesis on the full dataset.

14.1.4 RNA tertiary (3D) structure prediction: real tools and an honest scope decision

RNA 3D structure prediction has progressed substantially over the past two decades but has not reached anything resembling AlphaFold’s transformation of protein structure prediction. **FARFAR2** (Watkins, Rangan, & Das, 2020, building on Das & Baker’s 2007 original FARFAR), Rosetta’s fragment-assembly RNA 3D predictor, samples candidate tertiary folds by recombining short structural fragments drawn from known RNA structures and scoring them with a real, physics-informed energy function. **ARES** (Townshend et al., 2021) took a different, now-familiar approach — a real, rotation-equivariant graph neural network (the same $E(3)$ -equivariance principle Chapter 6 §6.4 introduced for small-molecule 3D networks) trained to re-score and rank FARFAR-generated candidate structures directly against real, experimental accuracy, rather than a physics-based energy alone. More recently, end-to-end deep-learning predictors have emerged directly in AlphaFold’s own methodological lineage: **RoseTTAFoldNA** (Baek et al., 2024), extending the RoseTTAFold protein-structure architecture to jointly model protein-nucleic-acid complexes; **trRosettaRNA** (Wang et al., 2023), a transformer-network RNA 3D predictor; and **DRfold** (Li et al., 2023), which integrates learned geometric potentials with end-to-end structure generation.

Despite this real progress, independent, blinded benchmarking — most visibly the RNA-Puzzles community trials and RNA’s first inclusion as a CASP15 prediction category — has consistently found RNA 3D structure prediction accuracy well behind protein structure prediction’s own CASP14/AlphaFold2-era benchmarks, for reasons intrinsic to the problem rather than to any one tool: RNA’s tertiary contacts (coaxial stacking, ribose-zipper and A-minor motifs, long-range tertiary base pairs) are individually weaker and more conformationally flexible than a folded protein’s hydrophobic core, and the training data available — the Protein Data Bank’s own RNA-only tertiary structures — is a much smaller, more repetitive corpus (dominated by rRNA, tRNA, and a handful of riboswitch/ribozyme families) than the protein structural universe Chapter 9’s AlphaFold2/ESMFold training sets drew on.

This chapter makes the same kind of honest, disclosed scoping decision Chapter 11 made for DiffDock (§11.2) and Chapter 13 made for its own library size (§13.3): none of FARFAR2, ARES, RoseTTAFoldNA, trRosettaRNA, or DRfold is a lightweight, pip-installable, CPU-tractable tool the way ESMFold’s single forward pass was for proteins in Chapter 9 — each requires either a full Rosetta installation and fragment-library infrastructure, or multi-gigabyte learned-model weights with a recommended GPU runtime, neither of which fits this book’s own free-tier-

Colab-compatible, pip-first authoring constraint. Section 14.3’s hands-on project therefore works entirely at the secondary-structure and duplex-thermodynamics level established in Sections 14.1.2-14.1.3, where ViennaRNA provides real, complete, lightweight, exactly-reproducible tools — the same “go where the real, tractable tooling is, and disclose why” principle this book has applied consistently since Chapter 11.

14.2 mRNA & siRNA Optimization

A therapeutic RNA sequence is rarely used exactly as first designed against its biological target; it is further optimized along three largely separable real axes — translation/expression (for a coding mRNA), stability, and immunogenicity — each with its own established design principles.

14.2.1 Codon optimization

The genetic code is degenerate: 61 sense codons encode only 20 amino acids, so a coding sequence for a fixed protein has enormous real sequence freedom at the nucleotide level, with no effect on the translated protein sequence itself. **Codon optimization** exploits that freedom, but not merely by maximizing the frequency-weighted “codon adaptation index” (CAI) against a host organism’s own codon usage table, as older heuristics did — Presnyak et al. (2015) established a real, mechanistic reason codon choice matters beyond translation speed alone: codon optimality is a major, direct determinant of **mRNA stability** itself, because slower-translating (non-optimal) codons trigger co-translational recruitment of the Ccr4-Not deadenylation complex, coupling translation elongation rate directly to mRNA decay. A therapeutic mRNA’s codon choice therefore trades off translation efficiency, mRNA half-life, and (Section 14.1.2) the resulting sequence’s own secondary-structure content, since synonymous codon substitutions change local base-pairing potential even though they don’t change the protein.

LinearDesign (Zhang et al., 2023) is a real, published algorithm that optimizes exactly this joint trade-off directly and efficiently: rather than optimizing codon choice and mRNA secondary structure as two separate steps, it formulates a single joint objective balancing translation efficiency (via CAI) against real, ViennaRNA-style minimum free energy (lower — i.e., more negative and more structured — being generally *less* desirable for coding-sequence stability and translation, the same “less intramolecular structure is generally more favorable” direction Section 14.1.2’s discussion already established for a guide strand’s own self-structure), and solves it with a real, polynomial-time lattice-parsing algorithm rather than a naive exponential search over the codon space. The paper reports this approach applied directly to real SARS-CoV-2 spike-protein mRNA vaccine design, reducing predicted mRNA structuredness and improving measured stability and expression relative to unoptimized or CAI-only-optimized sequences — a real, deployed-scale demonstration that the thermodynamic tools of Section 14.1 are not confined to siRNA design.

14.2.2 Stability enhancement

Beyond coding-sequence codon choice, an mRNA’s stability and translation efficiency depend heavily on its untranslated regions (UTRs), 5’ cap, and poly(A) tail — and, again, on secondary structure, now considered across the *entire* transcript rather than just the coding sequence. Leppek et al. (2022) ran a real, large-scale crowdsourced RNA design challenge (Eterna’s “OpenVaccine” project) that directly, empirically tested this relationship at scale: real, measured in-solution

stability across thousands of real, synthesized mRNA constructs correlated strongly with real, computed secondary-structure metrics, with unstructured 5' regions in particular associated with both higher stability and higher translation — the same accessibility logic Section 14.1.3 introduced for a *target* site applies, in a different but related sense, to a therapeutic mRNA's own 5' UTR as the site ribosomal scanning must traverse. This is a second, independent, real line of evidence (alongside LinearDesign) that the exact real ViennaRNA-computable quantities Section 14.1 introduces — MFE, base-pair probability, local structuredness — are not an academic exercise but a load-bearing part of how real, deployed RNA therapeutics are actually designed.

14.2.3 Immunogenicity reduction

Exogenous RNA delivered into a human cell is not immunologically inert: pattern-recognition receptors — endosomal Toll-like receptors TLR3, TLR7, and TLR8, and cytoplasmic sensors including RIG-I — evolved specifically to detect foreign RNA and trigger a real, potent innate immune (type I interferon) response, which for a therapeutic mRNA or siRNA is an unwanted side effect rather than the intended mechanism. Karikó, Buckstein, Ni, & Weissman (2005) established the real, foundational finding that this recognition is suppressed by naturally occurring nucleoside modifications: RNA synthesized with **pseudouridine** or **N1-methylpseudouridine** in place of standard uridine is dramatically less immunostimulatory via TLR3/7/8 while remaining translatable — the specific real chemical modification that underlies the mRNA vaccine platforms subsequently deployed at global scale, work recognized by the 2023 Nobel Prize in Physiology or Medicine. siRNA therapeutics address the analogous problem with a different, complementary set of real, established chemistries: 2'-O-methyl and 2'-fluoro ribose modifications and phosphorothioate backbone linkages, which simultaneously reduce innate-immune recognition and increase nuclease resistance. Delivery is a separate, real design axis on top of that chemistry: patisiran, the first FDA-approved siRNA drug, uses a lipid-nanoparticle carrier, while the liver-targeted siRNA drugs approved since (givosiran, lumasiran, inclisiran) instead conjugate the siRNA directly to N-acetylgalactosamine (GalNAc) for receptor-mediated hepatocyte uptake. None of these delivery/chemistry questions is addressed computationally in Section 14.3's hands-on project, which — like Chapter 11's docking-only scope or Chapter 12's ligand-alone MD scope — targets one real, well-defined, computationally tractable sub-problem (sequence-level knockdown efficacy) rather than the full therapeutic design pipeline.

14.3 Hands-on Project: Predicting siRNA Knockdown Efficacy from Real RNAi Data and Real RNA Thermodynamics

The project code lives in this chapter's folder (ch14_rna_therapeutics/). Given Section 14.1's toolbox and Section 14.2's design principles, this project builds a real, complete siRNA efficacy predictor: real measured silencing data in, real self-computed thermodynamic and sequence features, real trained models, real held-out evaluation.

Real data and its provenance, disclosed in full

The data. Huesken et al. (2005) profiled 19-mer siRNAs against a dual-luciferase reporter system in H1299 cells — the field's original large-scale siRNA efficacy dataset, and still the most widely used benchmark for this exact task two decades later (every rule-based and learned siRNA design

tool published since, including this chapter’s own, is implicitly or explicitly compared against it). The paper’s own original data hosting — Novartis’ BIOPREDsi web server — is no longer online, a real, disclosed constraint this chapter shares with several of Chapter 13’s own real 20-year-old fragment-screening data-access questions. `siRNA_efficiency.py` fetches the real sequences and real measured efficiency values live from `github.com/dimostzim/siRBench`, a small, actively maintained (as of this chapter’s writing) community redistribution assembled specifically because the original hosting disappeared; it carries no explicit open-source license. This chapter handles that honestly and conservatively rather than ignoring it: **only four raw, factual fields** are taken from that source — the 19-nt siRNA sequence, its 57-nt local target-mRNA context, the real measured efficiency value, and the source-study tag used to filter to the real Huesken-only subset (2,133 train + 228 test sequences, curated to zero sequence overlap between the two splits) — individual published scientific facts (a sequence, a number), not the redistribution’s own compiled/engineered feature columns, none of which this chapter reuses. Every feature this chapter’s own model trains on (below) is computed fresh, by this chapter’s own code, from those four raw fields, via ViennaRNA’s real thermodynamic engine.

A real data-quality check, not assumed. `curate_huesken_subset()` verifies, for every row, that the real 57-nt extended target context actually contains the real 19-nt target site at its documented [19:38] offset before accepting the row — a basic real consistency check against transcription/formatting errors in a third-party redistribution, following the same “verify, don’t trust, a bundled real dataset” discipline Chapter 13’s own replicate-well deduplication bug and fix established (§13.3).

Real, self-computed features

Fourteen real features are computed per sequence, none of them label-derived: sequence composition (overall GC content; seed-region [positions 2-8] GC content, the established RNAi “seed” window; 5’/3’ terminal base identity; a count of internal repeated 4-mers, Reynolds et al.’s 2004 internal-repeat liability); a compact, four-criterion `design_heuristic_score` — inspired by, but not a literal reproduction of, the qualitative design principles in Reynolds et al. (2004) and Ui-Tei et al. (2004), used here purely as this section’s rule-based baseline; and, from Section 14.1’s own real ViennaRNA machinery, the guide strand’s own self-structure MFE and ensemble diversity, the full siRNA-target duplex hybridization energy and its 5’-end-vs-3’-end terminal-stability asymmetry (`RNA.duplexfold`, the real, established biophysical basis of RISC strand-selection bias; Khvorova, Reynolds, & Jayasena, 2003; Schwarz et al., 2003), and the target site’s own real accessibility (unconstrained MFE and opening energy, Section 14.1.3).

```
def duplex_features(sirna: str, target_site: str) -> dict:
    full = RNA.duplexfold(sirna, target_site)
    five_prime = RNA.duplexfold(sirna[:6], target_site[-6:])
    three_prime = RNA.duplexfold(sirna[-6:], target_site[:6])
    return {
        "duplex_energy_total": round(full.energy, 3),
        "duplex_energy_5p_end": round(five_prime.energy, 3),
        "duplex_energy_3p_end": round(three_prime.energy, 3),
```

```

    "duplex_end_asymmetry": round(five_prime.energy - three_prime.energy, 3),
}

```

Real models and real, held-out evaluation

Four real predictors are trained on the real 2,133-sequence training split and evaluated, once, on siRBench’s own real, pre-defined, sequence-disjoint 228-sequence Huesken test split: the rule-based `design_heuristic_score` alone; a Random Forest (500 trees); an XGBoost gradient-boosted model; and a small PyTorch MLP (19→64→32→1, matching the same modeling family — a feed-forward neural network — Huesken et al. themselves used, on this chapter’s own, fully disclosed feature set rather than their original proprietary one), with early stopping on a 15% real validation split carved from the training data only, never touching the test set.

Model	Spearman ρ	Pearson r	R^2	RMSE	MAE
Rule-based score alone	0.267 (p = 4.5×10^{-5})	0.266	—	—	—
Random Forest	0.583 (p < 10^{-6})	0.567	0.316	0.131	0.102
XGBoost	0.573 (p < 10^{-6})	0.554	0.298	0.133	0.103
PyTorch MLP	0.543 (p < 10^{-6})	0.522	0.260	0.136	0.106

(R^2 /RMSE/MAE are omitted for the rule-based score: it is a bounded, 0-4 integer count, not a value calibrated to the real efficiency scale, so those absolute-error metrics are not meaningful for it — Spearman/Pearson rank/linear correlation are the real, comparable quantities across all four rows.)

Reading the real result. All three learned models comfortably and significantly outperform the rule-based baseline (Spearman ρ around 0.54-0.58 versus 0.267) — a real, substantial gain from letting a model learn the real, quantitative relationship between the thermodynamic and compositional features directly, rather than combining them through four hand-picked binary criteria. The **Random Forest is the strongest model on every real metric** measured, ahead of both XGBoost and the PyTorch MLP — a genuinely modest, believable margin (not one this chapter over-interprets as “tree ensembles are better than neural networks for this problem” from a single train/test split), consistent with the real, well-established finding in the broader siRNA-efficacy literature that this problem — a few dozen well-engineered biophysical/compositional features against a noisy experimental readout from a single assay platform — sits in a regime where gradient boosting and random forests are frequently competitive with or ahead of small neural networks, unlike the large-scale sequence/graph problems (Chapters 6-10) where deep learning’s advantage is much larger. An R^2 around 0.30 and Spearman ρ around 0.58 are real, modest, literature-consistent numbers for this exact task on this exact dataset — well above the rule-based floor and well above chance, but not the near-perfect predictor a compact, fourteen-feature model on a noisy single-cell-line luciferase-reporter assay was ever going to produce.

Feature importances confirm established biology, not an artifact. Random Forest’s single most important feature, by a wide margin, is `pos1_is_AU` (importance 0.28 of 1.0 total; XGBoost ranks the same feature even higher, at 0.45) — whether the guide strand’s 5′-terminal base is A or U. This is not a novel finding; it is this chapter’s own, independent, real quantitative confirmation of the single best-established qualitative rule in the RNAi design literature (Khvorova et al., 2003; Schwarz et al., 2003; Ui-Tei et al., 2004): a weakly paired (A/U) 5′ terminus lowers that end’s local duplex stability, biasing RISC to load the intended guide strand rather than the passenger strand as the functional antisense species. Seeing this exact, decades-old rule re-emerge as the dominant learned feature from a real, independent, from-scratch feature-engineering-and-training pipeline is a meaningful real sanity check on this section’s whole approach, in the same spirit as Chapter 11’s redocking-validation control or Chapter 13’s PAINS/QED sanity checks on drug-like chemistry: a model that had *not* rediscovered this relationship would be the result worth distrusting. The `duplex_energy_total` and `target_unconstrained_mfe/target_opening_energy` features — Section 14.1.3’s real accessibility machinery — cluster immediately behind it in both models’ importance rankings, a real, quantitative confirmation that target-site accessibility genuinely carries independent predictive signal on this dataset, not merely a theoretically motivated feature that turned out not to matter in practice.

Limitations and what comes next

This project builds a real, complete, held-out-evaluated siRNA efficacy predictor from real published silencing data and real, independently verifiable RNA thermodynamics — but several real, disclosed limitations bound how far its conclusions generalize. First, the Huesken dataset itself covers only 34 real target genes; because the redistributed data this chapter fetches carries no gene/transcript identifier, the sequence-disjoint train/test split used here cannot also be gene-disjoint, so some of the reported performance may reflect partial memorization of a given transcript’s own local structure/accessibility profile rather than purely sequence-generalizable signal — the same category of leakage concern Chapter 5’s scaffold split was built specifically to rule out for small-molecule QSAR, left here as an open, disclosed gap rather than a silently assumed non-issue. Second, every measurement in this dataset comes from a single assay platform (dual-luciferase reporter, H1299 cells); Section 14.3’s model has not been tested against knockdown measured by direct mRNA quantification, a different cell type, or a chemically modified (2′-OMe/2′-F, Section 14.2.3) siRNA, any of which could shift the real, learned feature-efficacy relationships. Third, the training data source itself is an unlicensed, community-maintained redistribution of a two-decade-old dataset — disclosed in full above, and mitigated by using only the raw, factual (sequence, efficiency) fields and independently verifying their internal consistency, but not eliminated as a provenance caveat. Chapter 15 shifts the book’s focus once more, from RNA’s own real biophysics to a different real cross-cutting question: how physical conservation laws and quantum-mechanical electronic-structure methods can be incorporated directly into the learned models Chapters 1-14 have each built from a different real angle.

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All feature-computation examples, model metrics, and feature-importance values cited in Section 14.3 were computed directly by running `sirna_efficacy.py` against the real, cached, curated Huesken train/test splits on 2026-08-21, not taken from a secondary source — see `results/sirna_efficacy_results.json` to reproduce.